Sorting

sort(arr.begin(), arr.end());

{1, 5, 2, 6, 3}

⇓

{1, 2, 3, 5, 6}

Ways of sorting

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

Bubble Sort

cold drink bubble

$i = 1$

0	1	2	3	4
5	6	2	3	1

$arr[j] < arr[j+1]$

swap (6,2)

0	1	2	3	4
5	2	6	3	1

swap (6,3)

0	1	2	3	4
5	2	3	6	1

swap (6,1)

0	1	2	3	4
5	2	3	1	6

$i = 2$

0	1	2	3	4
5	2	3	1	6

swap (2,5)

2	5	3	1	6

swap (5,3)

2	3	5	1	6

swap (5,1)

2	3	1	5	6

$i = 3$

0	1	2	3	4
2	3	1	5	6

$\xrightarrow{\text{swap}}$ $(3, 1)$

0	1	2	3	4
2	1	3	5	6

i j

$arr[j] < arr[j+1]$

$i = 4$

2	1	3	5	6

j

$\longrightarrow$

1	2	3	5	6

$i=0$

~~6~~ 1	~~2~~ 6	~~2~~ 6	~~3~~ 6

$i=4$

$\{1,2,3,6\}$

$n=5;\ i=1$

$j \in [0,3]$

$i=2;\ j \in [0,2]$

$i=3;\ j \in [0,1]$

$i=4;\ j \in [0,0]$

$j \in [0, n-i-1]$

```
n=arr.size();
for(int i=1; i<n; i++)
{
    bool swapped = false;
    for(int j=0; j<n-i-1; j++)
    {
        if (arr[j] < arr[j+1])
        {
            swap(arr[j], arr[j+1]);
            swapped=true;
        }
    }
    if (!swapped) break;
}
```

!false = true

Boolean ops:

True & False
(T) (F)

1. AND (&)
2. OR (|)
3. NOT (!)
(like complement for calculating opposite)

! True = False

! False = True

flag = false;

! flag → true

② Selection Sort

min.

$i=0$

0	1	2	3	4
5	6	2	3	1

i j j j j

(i+1) min=2 (4)

swap (arr[4], arr[i])

1	6	2	3	5

$i=1$

0	1	2	3	4
1	6	2	3	5

i j

min=2 (2)

swap (6,2)

1	2	6	3	5

$i=2$

0	1	2	3	4
1	2	6	3	5
		i	j	

$\xrightarrow{\text{swap}(6,3)}$

1	2	3	6	5

$i=3$

0	1	2	3	4
1	2	3	6	5
			i	j

$\xrightarrow{\text{swap}(6,5)}$

1	2	3	5	6

$n=5$

$i \in [0, n-2]$

$j \in [i+1, n-1]$

```
for (int i=0, i<n-2 ; i++){
    int min_idx=i;
    for(int j= i+1; j<n; j++){
        if (arr[j] < arr[min_idx])
            { min_idx= j ;
        }
    }
    swap(arr[i], arr[min_idx]);
}
```

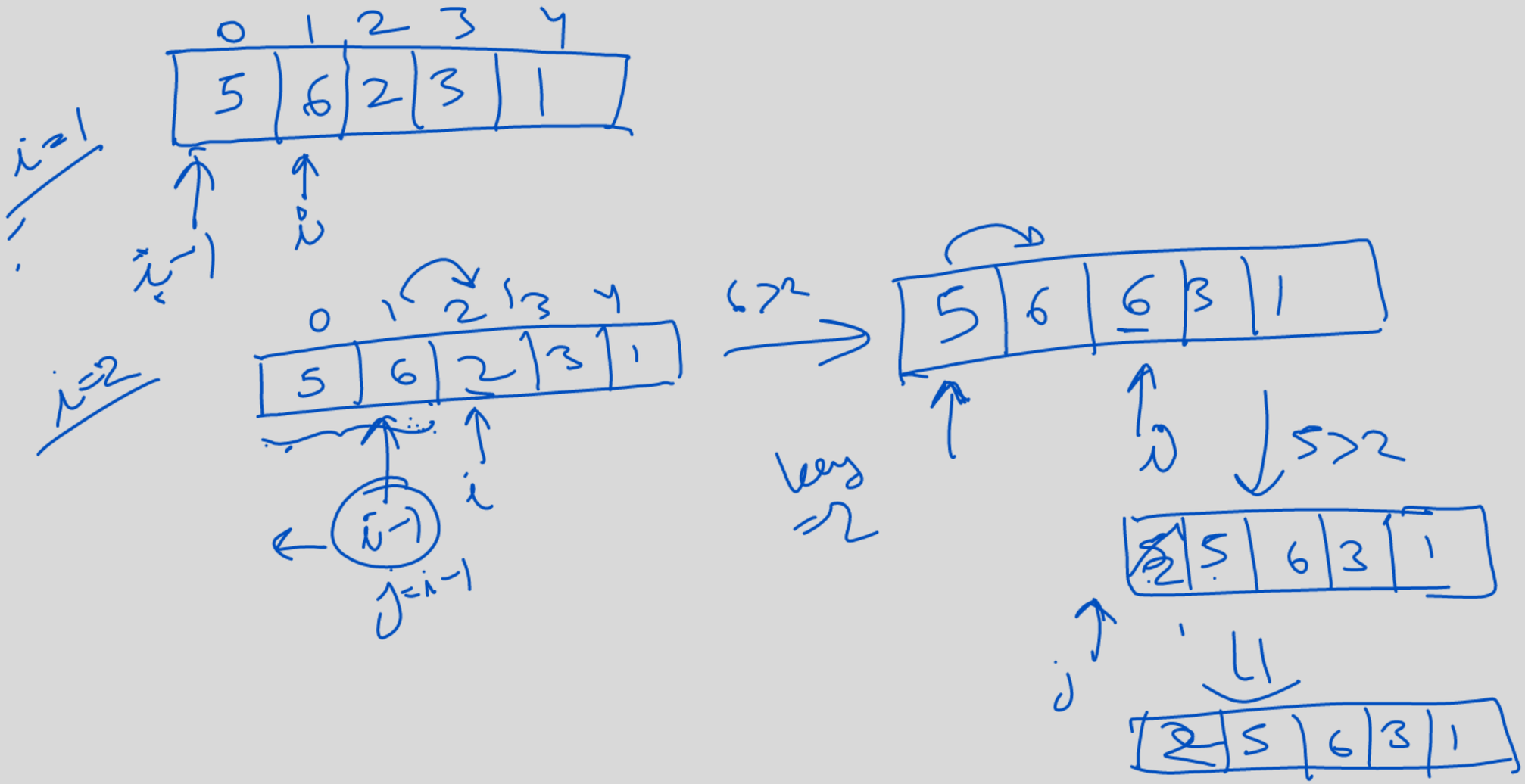
Insertion Sort

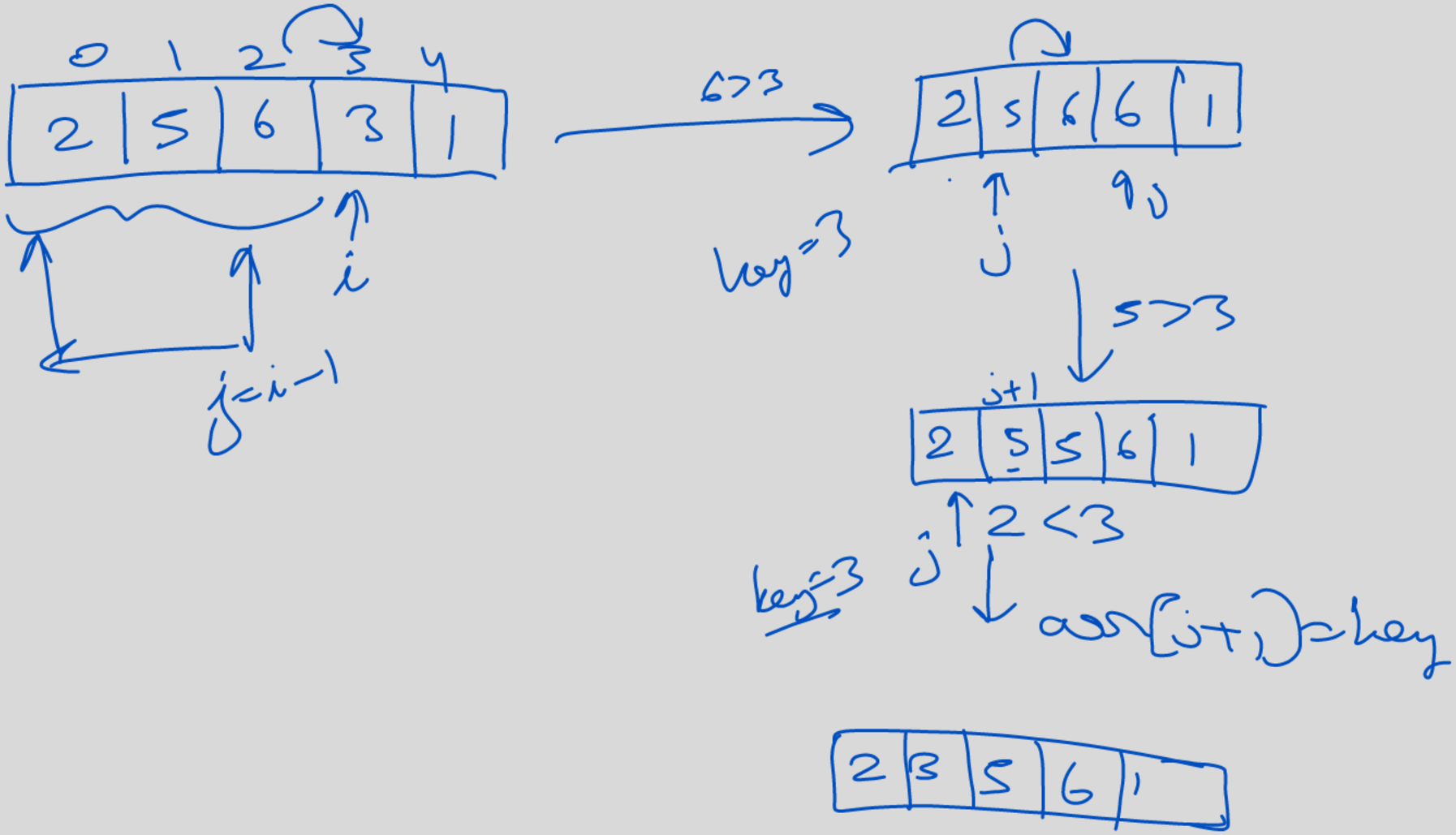
K Q J 10 9

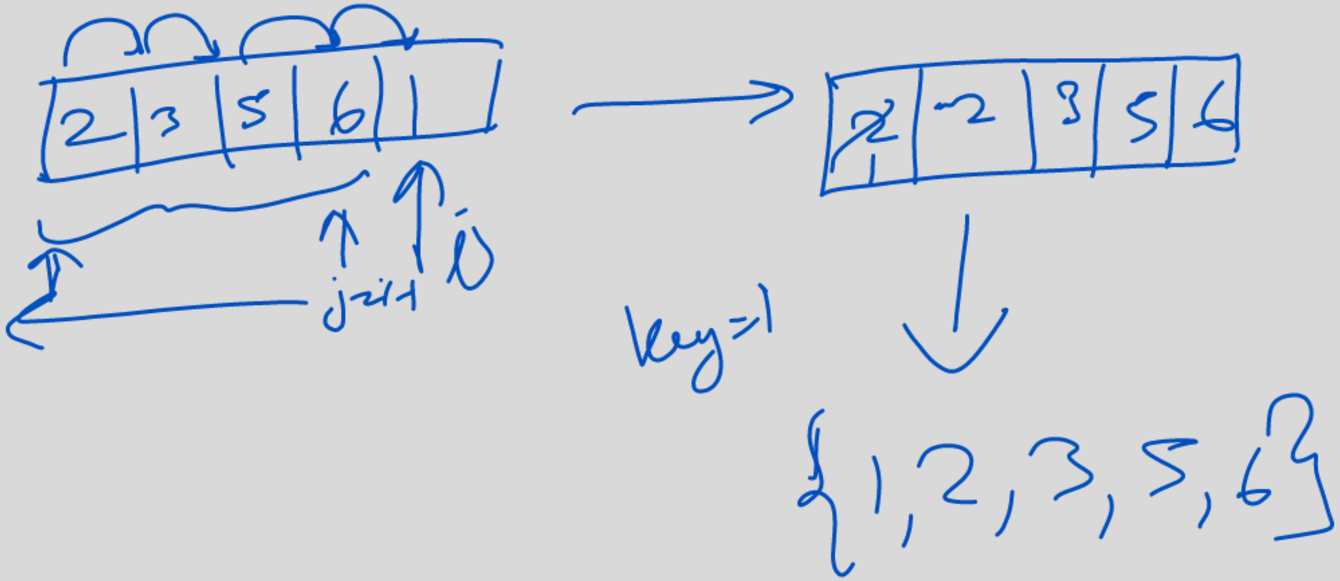
shuffle

K J 9 10 Q

Q 10


i=1
0 1 2 3 4
5 6 2 3 1
i-1
i
i=2
0 1 2 3 4
5 6 2 3 1
i-1
i
j=i-1
6>2
key =2
5 6 6 3 1
j
5>2
2 5 6 3 1
j
2 5 6 3 1


0 1 2 3 4
2 5 6 3 1
i
j=i-1
6>3
2 5 6 6 1
key=3
j
5>3
j+1
2 5 5 6 1
2<3
key=3
j
arr[j+1]=key
2 3 5 6 1


2 3 5 6 1
j=i+1
i
key=1
2 2 3 5 6
1
{1,2,3,5,6}

344 Reverse String

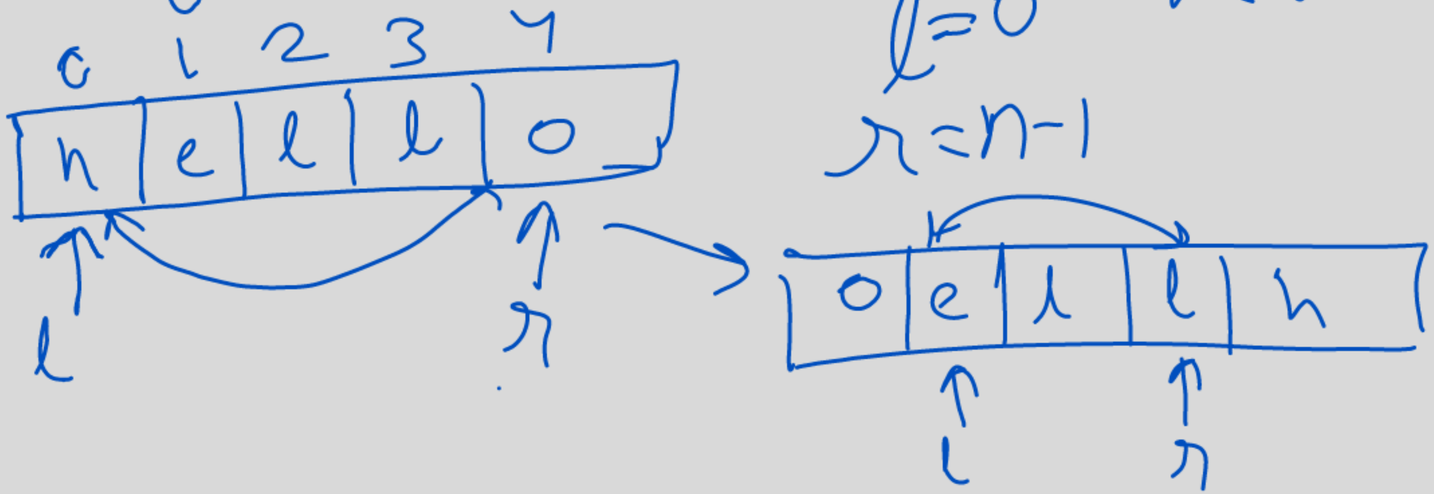
i/p: {"h", "e", "l", "l", "o"}

o/p: {"o", "l", "l", "e", "h"}

"Kaustav"

'K' {"abc", "def", "efg"}

$l = 0$ $r = n-1$ n = arr.size();



{o, l, l, e, h}

e r

$$\{h, e, l, l\} \rightarrow \{l, e, l, h\}$$

$$\{l, l, e, h\} \quad (l < r)$$

$$\{h, e, l, l\} \leftarrow \{l, e, l, h\}$$

189: Rotate Array

arr = {1, 2, 3, 4, 5, 6, 7}

k = 3

{7, 1, 2, 3, 4, 5, 6}

{6, 7, 1, 2, 3, 4, 5}

{5, 6, 7, 1, 2, 3, 4}

Brute force:

$\{1, 2, 3, 4, 5, 6, 7\}$

$\Downarrow$

$\{\cancel{7}^{7}, 1, 2, 3, 4, 5, 6\}$

key=7

key=6 $\{\cancel{7}^{6}, 7, 1, 2, 3, 4, 5\}$

key=5 $\{\cancel{5}^{5}, 6, 7, 1, 2, 3, 4\}$

k=7

$\{1, 2, 3, 4\}$ n=4

k=2

$\Downarrow$

$\{\cancel{3, 4, 1, 2}\}$

$\boxed{K=8}$

k=6 $\{3, 4, 1, 2\}$

k=7 $\{2, 3, 4, 1\}$

k=3 $\{2, 3, 4, 1\}$

k=8 $\{1, 2, 3, 4\}$

k=4 $\{1, 2, 3, 4\}$

k=5 $\{4, 1, 2, 3\}$

$\boxed{k = k \% n}$

$n=4$ $\{1,2,3,4\}$

$k=7$

$k=3$

$k=13$

$k=1$

$13 \cdot 1 \cdot 4$

$$\begin{array}{r} 3 \\ 4\overline{)13} \\ -12 \\ \hline 1 \end{array}$$

$3 \cdot 1 \cdot 4 = 3$

$k=3$

$$\begin{array}{r} 0 \\ 4\overline{)3} \\ -0 \\ \hline 3 \end{array}$$

$\{1,2,3,4,5,6\}$ $\underline{k=3}$

$n=6$ $rev(0,n-1)$

$\{4,5,6,1,2,3\}$

$\{\overset{0}{6},\overset{1}{5},\overset{2}{4},\overset{3}{3},\overset{4}{2},\overset{5}{1}\}$

$rev(0,k-1) = rev(0,2)$

$\{4,5,6,3,2,1\}$

$rev(k,n-1)$

$rev(3,5)$

$\{4,5,6,1,2,3\}$